■ SHOULDER AND ELBOW

Temporal trends and geographical variation in the use of subacromial decompression and rotator cuff repair of the shoulder in England

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We explored the trends over time and the geographical variation in the use of subacromial decompression and rotator cuff repair in 152 local health areas (Primary Care Trusts) across England. The diagnostic and procedure codes of patients undergoing certain elective shoulder operations between 2000/2001 and 2009/2010 were extracted from the Hospital Episode Statistics database. They were grouped as 1) subacromial decompression only, 2) subacromial decompression with rotator cuff repair, and 3) rotator cuff repair only.

The number of patients undergoing subacromial decompression alone rose by 746.4% from 2523 in 2000/2001 (5.2/100 000 (95% confidence interval (CI) 5.0 to 5.4) to 21 355 in 2009/2010 (40.2/100 000 (95% CI 39.7 to 40.8)). Operations for rotator cuff repair alone peaked in 2008/2009 (4.7/100 000 (95% CI 4.5 to 4.8)) and declined considerably in 2009/2010 (2.6/100 000 (95% CI 2.5 to 2.7)).

Given the lack of evidence for the effectiveness of these operations and the significant increase in the number of procedures being performed in England and elsewhere, there is an urgent need for well-designed clinical trials to determine evidence of clinical effectiveness.

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The prevalence of shoulder conditions is estimated to be between 16% and 26% in the United Kingdom, with around 1% of adults consulting their general practitioner (GP) each year regarding a first episode of shoulder pain.[1,2] Shoulder problems are responsible for 2.4% of all United Kingdom GP consultations, and their prevalence increases with age and is greater in women.[3] Conditions affecting the rotator cuff tendons and the subacromial bursa account for up to 70% of shoulder problems.[4-6] The annual financial burden of managing shoulder pain in the USA has been estimated to be US$3 billion,[7] and it accounts for 4.5 million visits to physicians annually.[8]

A mechanical cause for the pain is thought to include impingement between the tendons of the rotator cuff and the overlying acromion and coracohumeral ligament. Progressive changes may occur, with the acromion developing a bony spur at its anteroinferior margin. This makes physical contact more likely, particularly in overhead positions of the arm. As part of this process both the subacromial bursa and the adjacent tendons may become inflamed and swollen.

Many patients with shoulder pain will respond to conservative treatment, including physiotherapy and steroid injections.[9] Severe pain combined with functional restriction that fails to improve with conservative measures is an indication for surgery.[4] The most common surgical procedure is arthroscopic subacromial decompression, involving removal of the spur, release of the coracohumeral ligament and excision of bursal tissue.[4] Rotator cuff repair, usually in conjunction with decompression is performed if the tendons have become detached from their insertion on to the humerus.

Evidence supporting the effectiveness of this surgery is not clear, and some reports show that it may be no more beneficial than physiotherapy.[10-12] Supervised exercises delivered to patients on a waiting list for subacromial decompression can significantly improve symptoms and reduce the number of those who ultimately proceed to surgery.[13] Assessment of the cost of treatment suggests that the addition of surgery, compared with physiotherapy alone, is not cost-effective.[14] There are > 300 000 rotator cuff repairs performed annually in the United States, although there is little evidence for its effectiveness and cost-effectiveness.[10-12]

We have investigated the trends over time of the provision of subacromial decompression and rotator cuff repair in England. We also explored national geographical variations in the provision of these procedures to determine whether any changes in rates of surgery occur uniformly across the population.

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Patients and Methods

Data were obtained from the Hospital Episode Statistics (HES) database, which holds information on patients admitted to NHS hospitals in England, either as day cases or as ordinary admissions, and care given to private patients in NHS hospitals. Each record in the database relates to one so-called 'finished consultant episode'. This is the period of time an individual spends under the care of one NHS consultant. HES includes private patients treated in NHS hospitals, patients resident outside of England and care delivered by treatment centres (including those in the independent sector) funded by the NHS. The information held in the HES database includes the age and gender of patients, their area of residence, the reason for admission (diagnosis) to hospital, and the procedure that was undertaken. HES data for this analysis were extracted using the HES online service. Further information on HES is available from http://www.hesonline.nhs.uk.

Anonymised records for patients undergoing the following elective operations between 1 April 2000 and 31 March 2010 were extracted: 1) subacromial decompression only; 2) subacromial decompression and rotator cuff repair; and 3) rotator cuff repair only. These episodes were identified using a combination of the Classification of Surgical Operations and Procedures (OPCS-4.5)[15] codes in the procedure fields that contain information about a patient's operations, and the International Statistical Classification of Diseases, Injuries and Causes of Death (ICD-10)[16] codes from the diagnostic fields that contain information about a patient's illness or condition (see supplementary information for the full list of OPCS4 and ICD-10 codes used).

Statistical analysis. Stata v12.1 (StataCorp, College Station, Texas) was used for all statistical analyses. In order to describe trends in rates of admission over time between the financial years 2000/2001 and 2009/2010 for each of the three groups of procedure, we used direct standardisation using the 2002 England age/gender population to calculate age- and gender-standardised admission rates and confidence intervals (CI).[17] For the denominator, population counts by ten-year age groups, gender and year were obtained from the Office for National Statistics population estimates for England.[17] The Association of Public Health Observatories (APHO) tool was used to calculate the standardised rates to conform to the APHO standard.[18]

In order to explore geographical variation in the use of these procedures we identified the Primary Care Trust (PCT) area of residence of each patient. PCTs are the public authorities in England responsible for planning, securing, funding and coordinating the NHS services in a defined geographical area. They make decisions about how resources are invested, and what health services are provided at a local level. Since 2006 there have been 152 PCTs with an average population of 300 000. All PCT data are configured to the 2010 PCT boundaries so that they are consistent over time. For the calculation of rates, the numerator is the number of procedures within each age, gender, PCT group and year of admission.

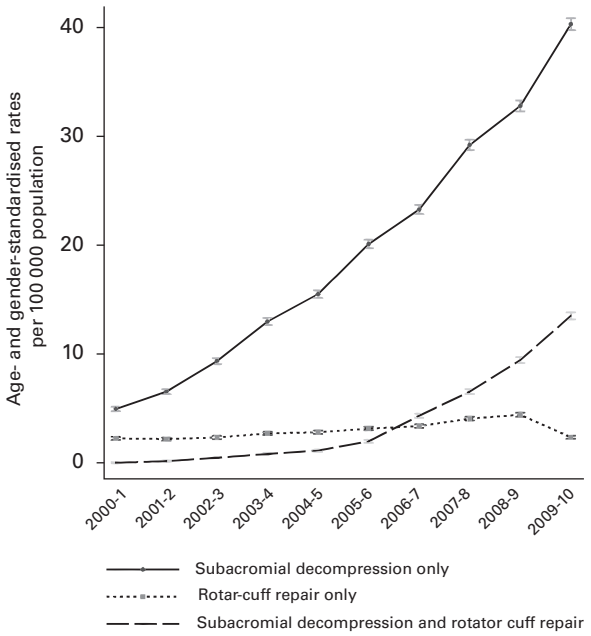

Fig. 1

Graph showing the age- and gender-standardised rates by type of procedure between 2000/1 and 2009/10.

For the denominator, PCT population counts were obtained from the National Centre for Health Outcomes Development (NCHOD) and configured to 2010 boundaries. Rates where the total number of events is less than six have been suppressed. The Geographic Information Systems (GIS) software MapInfo Professional 11 (Pitney Bowes Inc., Stamford, Connecticut, United States) was used to display maps of age- and gender-standardised procedure rates within each PCT.

Results

Between 1 April 2000 and 31 March 2010 a total of 101 254 patients underwent subacromial decompression in England (Group 1). Men accounted for 49.5% (n = 50 077) of procedures, women for 50.5% (n = 51 177). The mean age of admission was 54.8 years (SD 12.8) (16 to 92) for men and a mean of 55.3 years (SD 12.5) (11 to 92) for women. The number of procedures rose by 746.4%, from 2523 patients in 2000/2001 (5.2 per 100 000 (95% CI 5.0 to 5.4)) to 21 355 in 2009/2010 (40.2 per 100 000 (95% CI 39.7 to 40.8)). Trends in admission rates are shown in Figure 1, where it can be seen that there has been a dramatic increase in subacromial decompressive surgery. The numbers of patients receiving both subacromial decompression and rotator cuff repair (Group 2) remained low up until the year 2004/2005, after which point they increased substantially from 713 (1.4 per 100 000 (95% CI 1.3 to 1.5)) to 7330 procedures (13.7 per 100 000 (95% CI 13.4 to 14.0)) in 2009/2010.

Operations for rotator cuff repair only (Group 3) peaked in 2008/2009 at 2468 (4.7 per 100 000 (95% CI 4.5 to 4.8)), then declined considerably in 2009/2010 to 1399

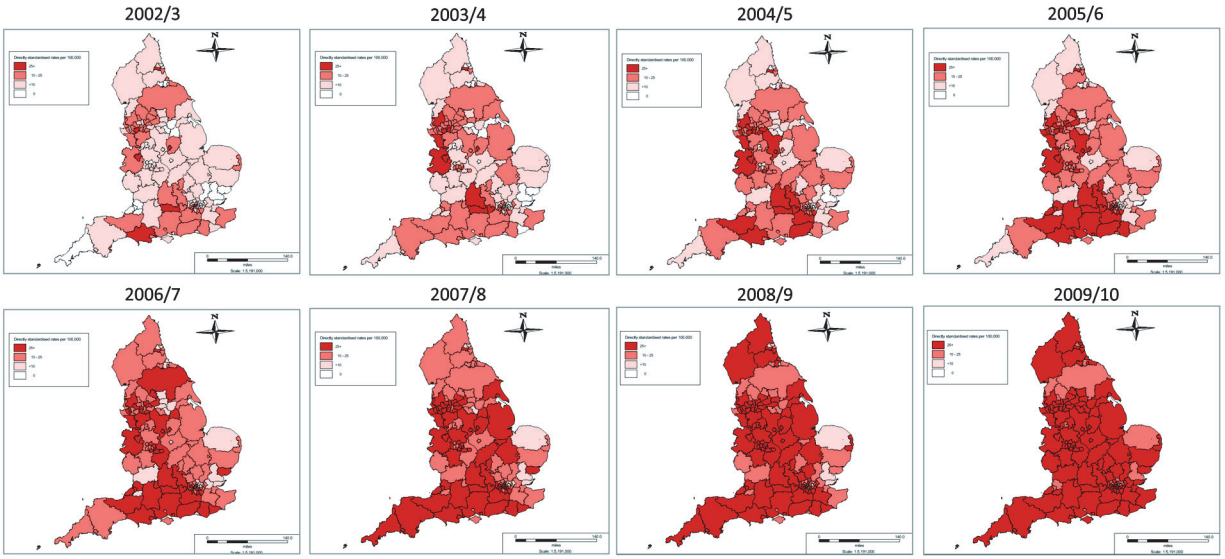

Fig. 2

Age- and gender-standardised rates of subacromial decompression alone by English Primary Care Trust between 2002/3 and 2009/10. The four colours represent the age- and gender-standardised rate of SAD only per 100 000 population in each PCT area: white, 0 per 100 000; light red, < 10 per 100 000; medium red, < 25 per 100 000; dark red, 25+ per 100 000.

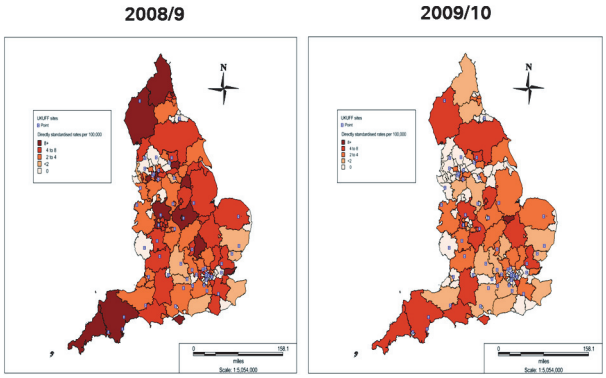

Fig. 3

Age- and gender-standardised rates of rotator cuff repair alone by English Primary Care Trust between 2008/9 and 2009/10. The five colours represent the age- and gender-standardised rate of RCR only per 100 000 population in each PCT area: white, 0 per 100 000; light orange, < 2 per 100 000; medium orange, 2 to 4 per 100 000; dark orange, 4 to 8 per 100 000; very dark orange, 8+ per 100 000. The UKUFF sites are marked with a hospital symbol.

(2.6 per 100 000 (95% CI 2.5 to 2.7)). We obtained additional HES data for subsequent years showing that this rate continued to fall in 2010/2011 to 1308 procedures (2.4 per 100 000 (95% CI 2.3 to 2.5)) and in 2011/2012 to 1110 procedures (2.0 per 100 000 (95% CI 1.9 to 2.1)).

Standardised rates of subacromial decompression only (Group 1) varied geographically according to the PCT (Fig. 2). In 2002/2003 only seven PCTs were performing a high number of these procedures (defined as an age- and gender-standardised rate ≥ 25 per 100 000 population). Increasing use of this procedure spread geographically across the country from these PCTs, such that by 2009/2010 83.6% of PCTs in England (127 of 152) were performing ≥ 25 procedures per 100 000 population.

Figure 3 shows rates of rotator cuff repair only (Group 3) by PCT in the year 2008/2009, when rates were at their highest, *versus* 2009/2010 when rates declined. The figure demonstrates substantial geographical variation in the use of this procedure. Whereas the overall rate in England was 4.7 per 100 000 in year 2008/2009, across PCTs this ranged from no procedures (or data suppressed as too few events) up to 17.7 per 100 000. The decline occurred uniformly across all areas of the country in 2009/2010.

Discussion

This study has demonstrated a substantial increase in the use of subacromial decompression over the past decade in England, with intervention rates increasing by 746.4% from 5.2 per 100 000 in 2000/2001 to 40.2 per 100 000 in 2009/2010. In 2002/2003 only a few PCTs were performing high numbers of these procedures, but over time increasing use of this procedure has spread across the country. The increasing use of this operation may be due to the greater awareness among patients and GPs of its availability. There has also been an increase in the numbers of specialist shoulder surgeons trained to perform these procedures, as determined by membership of the British Elbow and Shoulder Society (BESS), which showed an increase of 164% during this time.

Rotator cuff repair used in conjunction with subacromial decompression increased substantially from 2004/2005 onward, which may reflect a tendency of surgeons to use

less-invasive arthroscopic techniques. Over the last decade there has been uncertainty about which patients should be offered rotator cuff tear repair, and this may explain some of the trends we observed. It is possible that shoulder surgeons started to use subacromial decompression with rotator cuff repair at around this time owing to a perception that the two procedures should be combined, although there is no evidence to support this.

Rates of use of rotator cuff repair alone also increased steadily over time, peaking in 2008/2009 before falling away dramatically by 2009/2010, returning to rates for this type surgery seen ten years earlier. A potential explanation for the observed reduction is that it coincides with the start of the United Kingdom Rotator Cuff (UKUFF) Trial, which is a multicentre randomised controlled trial involving 58 surgical units and 90 surgeons from around the UK, measuring the clinical outcomes and cost effectiveness of different types of surgery for rotator cuff repair and comparing arthroscopic to open/mini-open repair. Assessing data from the follow-up questionnaire from the UKUFF surgeons suggests that attitudes to rotator cuff surgery have changed. It may be that simply running a large trial creates a change in practice even before the results are known.

In order to explore this further, the locations of centres in the UKUFF trial were overlaid on maps comparing rates of rotator cuff repair only by PCT in the year 2008/2009 compared with 2009/2010 (Fig. 3). The UKUFF centres are spread across the country and the decline in rates of rotator cuff repair only occurred uniformly across all areas of the country in 2009/2010. These rates appeared to decline notably in areas clustered around UKUFF sites, particularly sites with higher rates of surgery in 2008/2009.

An alternative explanation is that in 2009/2010 a dedicated subacromial decompression OPCS procedure code (O29.1) was introduced at the time of the observed decrease in rotator cuff repair. It is possible that the decrease in rotator cuff repairs is an artefact of a coding system that did not allow for the precise coding of subacromial decompression, and consequently some subacromial decompressions were incorrectly coded as rotator cuff repairs. We extracted additional data for the years 2010/2011 and 2011/2012, when rates of subacromial decompression continued to increase and rates of rotator cuff repair only continued to decline, and the use of code O29.1 accounted for only 10% of admissions in each year and has not been increasingly used. As the number of rotator cuff repairs per year is relatively small, the incorrect coding of a small percentage of subacromial decompressions, despite having no discernible effect on the trend of subacromial decompressions, could have a significant impact on the trend of rotator cuff repairs. However, the continuing increase in the numbers of subacromial decompressions after 2009/2010, combined with the continued decrease in rotator cuff repairs, suggests that this is unlikely to be a coding issue.

Increases in the provision of subacromial decompression are reflected in other healthcare systems, and recent reports from the United States describe a 254% increase from 29.9 to 102.2 per 100 000 in New York State (NYS) between 1996 and 2006, with only a 78% increase in orthopaedic surgery overall.[19] The NYS surgeons perform over five times as many decompressions *per capita* as do English surgeons. Analysis of data from the National Hospital Discharge survey in the United States between 1996 and 2006 showed that rotator cuff repairs increased by 141%, particularly for arthroscopically assisted repair (600%) compared with open repair (34%).[20] Significant geographical variation was shown in this study. The introduction of less-invasive arthroscopic techniques with lower risks probably accounts for some of the overall increase in the rate of surgery, but does not explain the geographical variation. Patient and disease characteristics have not changed over time, and there is a growing concern that these procedures are being over-used.[11,13]

Observational studies of patients undergoing subacromial decompression show good results in terms of reduced pain and better function, with high rates of satisfaction.[4,21-23] However, equally good outcomes have been found in studies following patients who had arthroscopic rotator cuff debridement or open repair in the absence of a subacromial decompression.[24] Comparative studies of subacromial decompression *versus* non-operative treatment, such as physiotherapy, have not shown any significant difference in outcome between the two treatment modalities.[10,11] There is a growing number of studies that have tried to assess the effectiveness of subacromial decompression against a control group.[14,24] In a recent randomised controlled trial, subacromial decompression plus bursectomy was compared with bursectomy alone and there was no significant difference in clinical outcome between the groups.[25] This finding suggests that removing acromial spurs might not be necessary, thereby challenging the principal indication for the procedure.

It is known that the very process of undergoing surgery can generate a beneficial outcome. This has been shown in the knee, where a placebo-controlled trial of arthroscopic surgery for osteoarthritis of the knee found that, despite early benefit, the results one year after arthroscopic lavage or debridement were no better than after an entirely placebo operative procedure.[26]

The strength of this study includes the use of a large nationally representative dataset of all admissions for subacromial decompression and rotator cuff repair in NHS hospitals across England. All PCT data were configured to the 2010 boundaries, so are consistent over time, thereby overcoming boundary changes. Limitations are that routine hospital admission data are collected for administrative rather than research purposes and concerns have been raised over the completeness and accuracy of such data. A weakness of using HES data is that private operations are not included. Hence we will have underestimated the

overall numbers of procedures being undertaken. An additional weakness is that we are unable to distinguish open procedures from arthroscopic or arthroscopically assisted procedures. A review by the private health insurance company BUPA (British United Provident Association) of the rates of extensive open repair of the rotator cuff and arthroscopic subacromial decompression found that those with private insurance were more than three times more likely to have these procedures than NHS patients.

It should also be noted that operation rates are adjusted only by age and gender, and the observed geographical variation in rates of surgery may be partly explained by adjusting for other geographical confounding factors. It is conceivable that during the study period the quality of recording of these procedures improved, thereby exaggerating any increase in incidence.

Despite the lack of evidence for the effectiveness of subacromial decompression and rotator cuff repair surgery, there has been a significant increase in the numbers of these procedures being performed in England. This increase is widespread across PCTs in England and is not just limited to one region. These findings highlight the importance of monitoring the implementation of new unproven surgical techniques, and the urgent need for well-designed clinical trials.

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